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Table 2. Water soluble vitamins		
Vitamins	**Functions**	**Symptoms of Deficiency**
B_1 (Thiamine)	Involved in carbohydrate metabolism. Act as a coenzyme	**Beriberi:** affects muscular, nervous and cardiovascular system
B_2 (Riboflavin)	Acts as coenzyme in oxidation and reduction reactions	Inflammation, soreness and fissures in the corners of the mouth, lips and tongue. Loss of appetite. Skin and eye disorder.
B_3 (Pantothenic acid)	Acts as coenzyme A and is essential for the metabolism of fats and carbohydrates	Gastrointestinal disorders, anaemia, Burning feet syndrome, etc.
B_4 (choline)	Precursor for acetylcholine	Fatty liver.
B_5 (Niacin / Nicotinic acid)	Derivatives of coenzymes	**Pellagra** (4D Syndrome) characterised by dermatitis, diarrhoea and dementia (mental deterioration) and death.
B_6 (Pyridoxine)	Haemoglobin formation, brain, heart and liver activities	**Dermatitis,** convulsions, muscular twitching and **anaemia**
B_7 (Biotin) / Vit.H	Acts as a coenzyme in synthesis of fat, glycogen and amino acids	**Dermatitis**
B_9 (Folic acid)	It acts as a co-enzyme for synthesis of nucleic acid and essential for growth and formation of RBC	**Megaloblastic anaemia** (large, immature, nucleated RBC in blood)
B_{12} (Cobalamine)	Promotes DNA synthesis. Necessary for maturation of RBC and formation of myelin sheath.	**Pernicious anaemia** (immature nucleated RBC without haemoglobin). Causes nervous disorder.
C (Ascorbic acid)	Acts as an antioxidant. Strengthens the immune system. Necessary for healthy gums and teeth.	**Scurvy** (Sailor's disease) characterized by spongy and bleeding gums, falling of teeth, fragile bones, delayed wound healing etc. - Infantile scurvy)

Food adulterants cause harmful effects in the form of headaches, palpitations, allergies, cancers and in addition reduces the quality of food. Common adulterants are addition of citric acid to lemon juice, papaya seeds to pepper, melamine to milk, vanillin for natural vanillin, red dyes to chillis, lead chromate and lead tetraoxide to turmeric powder, etc.,